



Updating JUnit Dependencies in Gradle

A practical guide to managing **JUnit 6 dependencies** in your **Gradle projects**



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Updating JUnit Dependencies in Gradle

- **JUnit configuration** is done inside the `build.gradle` file
- The **default template** may include **older versions** of **JUnit** dependencies
- You can update them using **Maven Repository** as the source of **latest versions**
- Search "**JUnit Jupiter**" to locate the correct dependencies
- Replace **outdated entries** under `testImplementation` or `testRuntimeOnly`



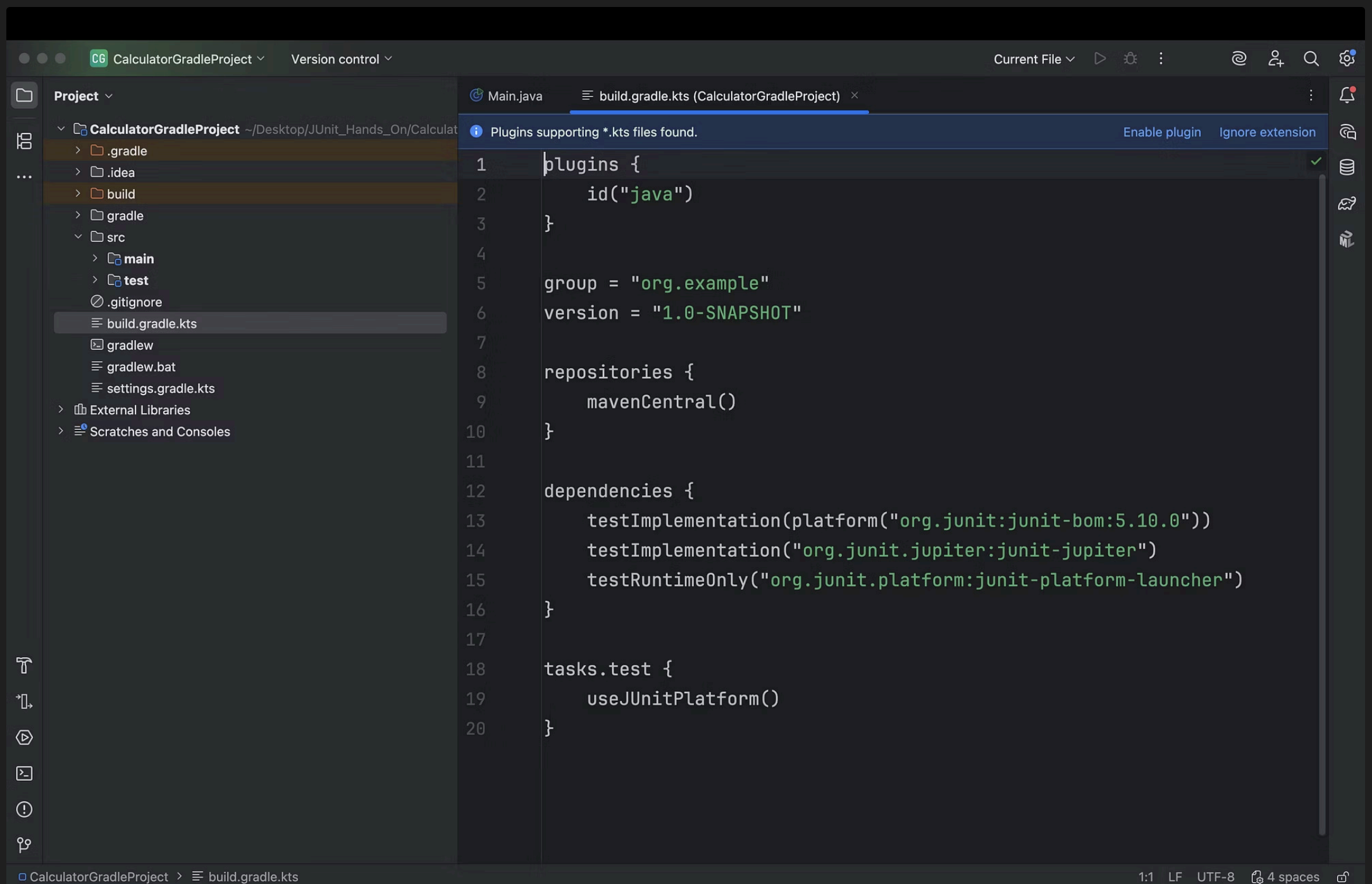
Enabling a plugin to support
.kts files

Enabling a plugin to support .kts files

You can **skip** this and the **next two slides** if you do not receive the message **Plugins supporting *.kts files found.**

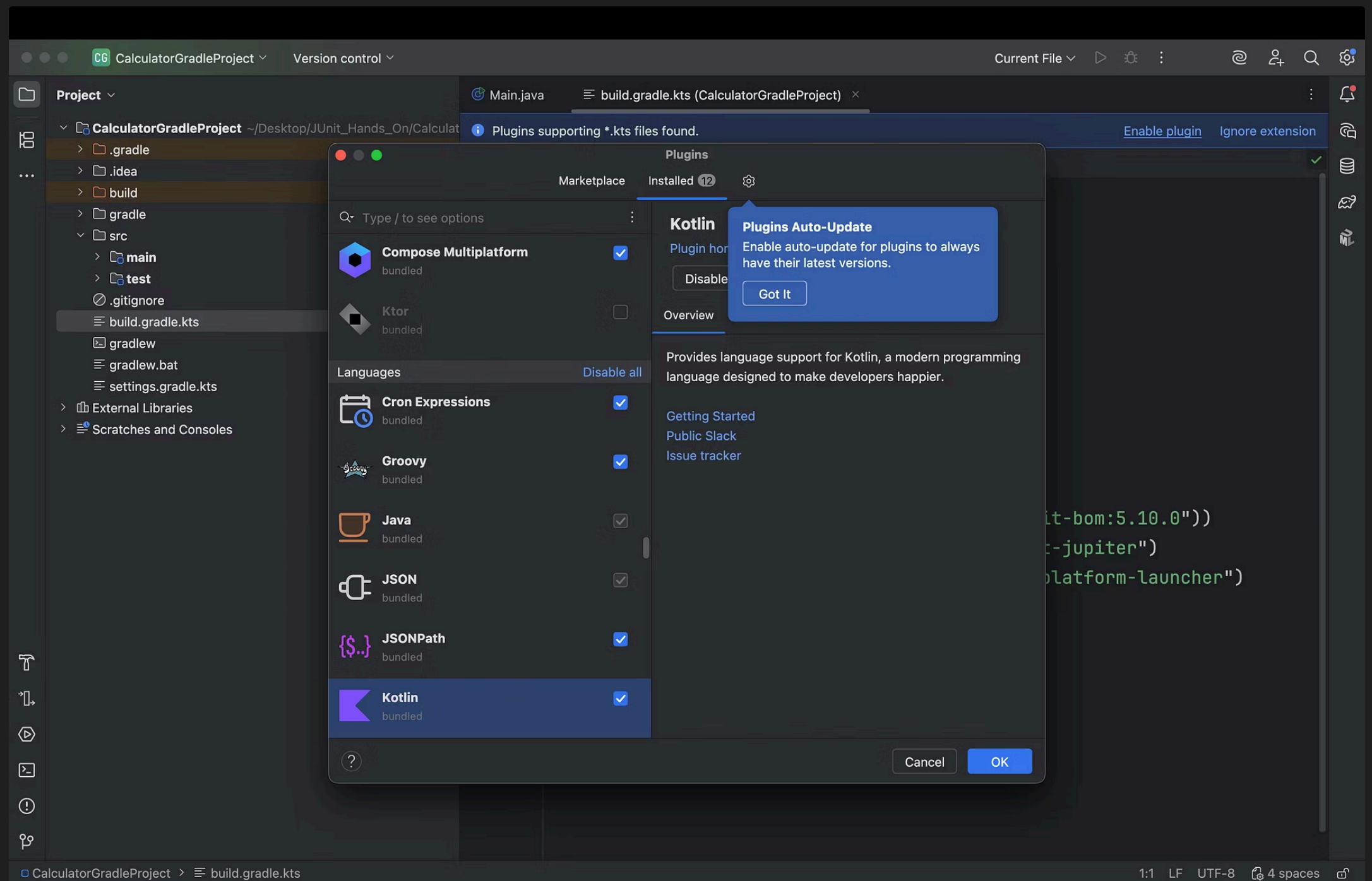
In my **local installation**, I got a message that I can enable a plugin to support `.kts` files.

Let's enable it!



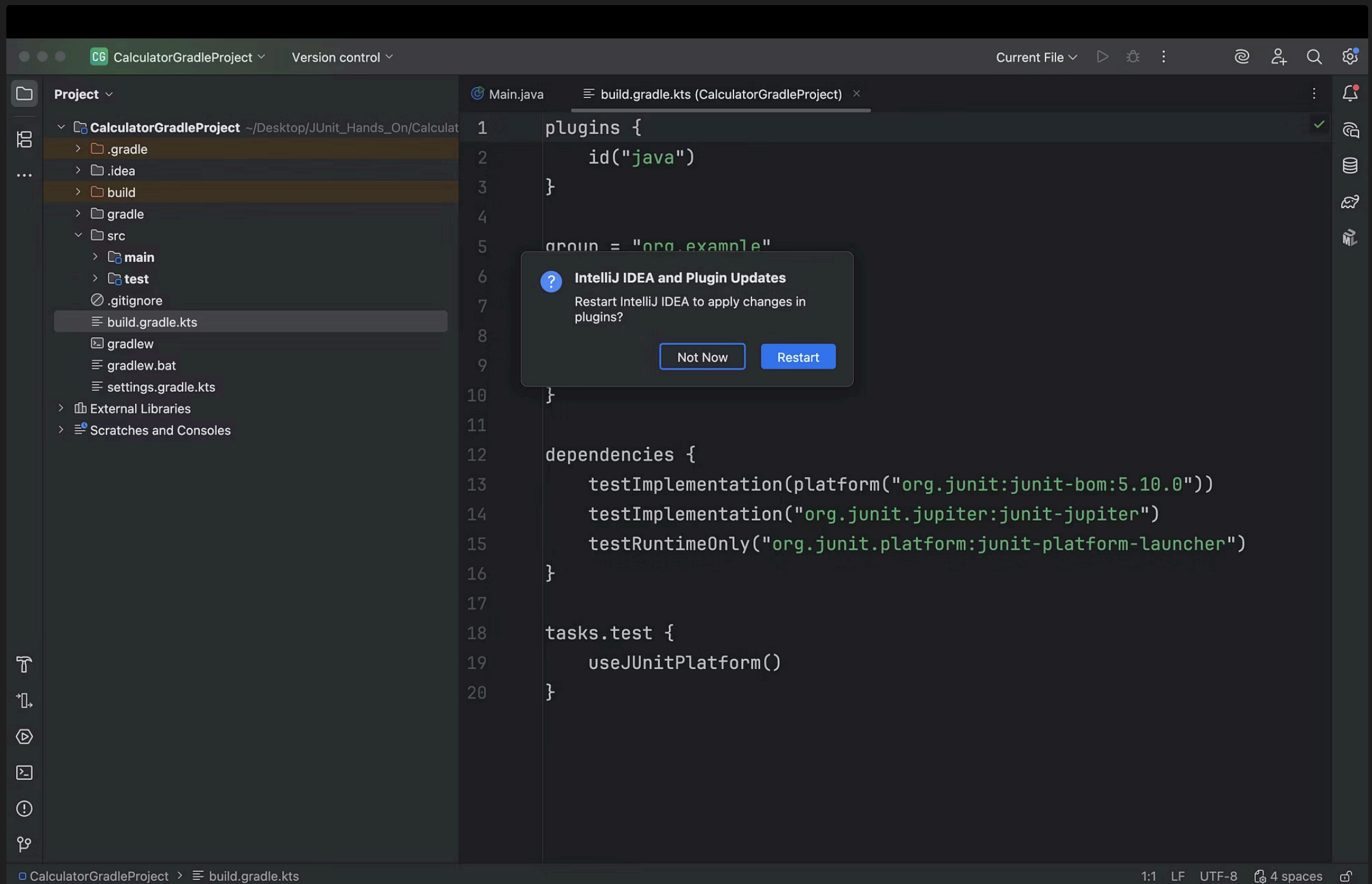
Enabling a plugin to support .kts files (contd.)

Install the Kotlin plugin



Enabling a plugin to support .kts files (contd.)

Restart IntelliJ



Updating the default JUnit dependencies in Gradle

Updating Individual Dependencies



Update iunit-jupiter-api and junit-jupiter-engine to desired versions



Use the Gradle Short format from Maven Repository



Additional module: junit-jupiter-params for parameterized tests

These **can** all be **added manually one by one** to fine-tune project setup

Gradle will **download dependencies** after the project is **refreshed**

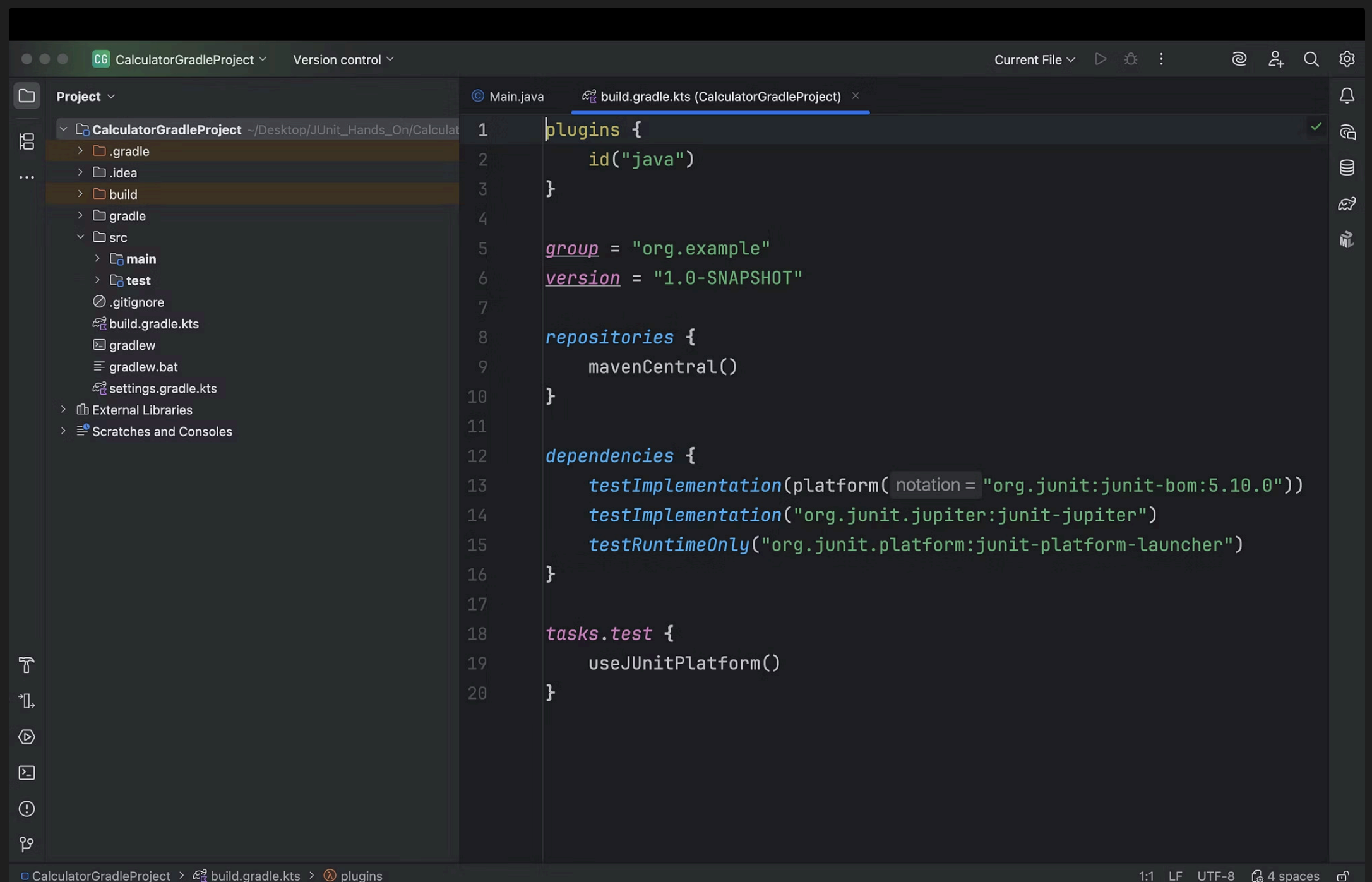


Getting Started

Updating Individual Dependencies (contd.)

Notice that the **default** version of **JUnit** when creating a **Gradle** project is **5.10.0**.

Let's **update** this version to the newer **6** version.



```
1 plugins {
2     id("java")
3 }
4
5 group = "org.example"
6 version = "1.0-SNAPSHOT"
7
8 repositories {
9     mavenCentral()
10 }
11
12 dependencies {
13     testImplementation(platform("org.junit:junit-bom:5.10.0"))
14     testImplementation("org.junit.jupiter:junit-jupiter")
15     testRuntimeOnly("org.junit.platform:junit-platform-launcher")
16 }
17
18 tasks.test {
19     useJUnitPlatform()
20 }
```

Getting Started

Updating Individual Dependencies (contd.)

Maven repository → Jupiter (individually) → dependencies section of build.gradle → reload project.

The screenshot shows the Maven Repository search results for the query 'jupiter'. The page is divided into three main sections: Repository, Group, and Category. The Repository section lists various repositories with their respective counts. The Group section lists various groups with their respective counts. The Category section is currently empty. The main content area displays the search results, showing 215 results found. The results are sorted by Relevance. The top four results are listed below:

Rank	Artifact Name	Usage Count	License
1	JUnit Jupiter API org.junit.jupiter » junit-jupiter-api	20,619 usages	EPL
2	JUnit Jupiter Engine org.junit.jupiter » junit-jupiter-engine	19,511 usages	EPL
3	JUnit Jupiter (Aggregator) org.junit.jupiter » junit-jupiter	13,153 usages	EPL
4	JUnit Jupiter Params org.junit.jupiter » junit-jupiter-params	8,372 usages	EPL

Each result includes a brief description and the last release date (Oct 31, 2025).

Getting Started

Updating Individual Dependencies (contd.)

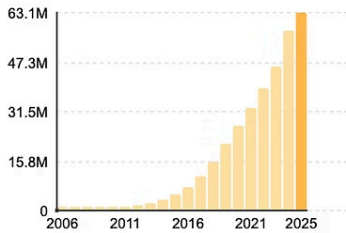
← → ↻ 🔍 mvnrepository.com/artifact/org.junit.jupiter/junit-jupiter-api/6.1.0-M1

MVN REPOSITORY

Search for groups, artifacts, categories

Search

Indexed Artifacts (63.1M)



Year	Indexed Artifacts (M)
2006	0
2011	0.5
2016	1.5
2021	15.8
2025	63.1

Popular Categories

Testing Frameworks & Tools

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Core Utilities

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Annotation Libraries

HTTP Clients

Language Runtime

Logging Bridges

Dependency Injection

XML Processing

Home » org.junit.jupiter » junit-jupiter-api » 6.1.0-M1



JUnit Jupiter API » 6.1.0-M1
JUnit Jupiter is the API for writing tests using JUnit 5.

License	EPL 2.0
Categories	Testing Frameworks & Tools
Tags	quality junit testing api
HomePage	https://junit.org/ Inspect URL
Date	Nov 17, 2025
Files	pom (3 KB) jar (252 KB) View All
Repositories	Central
Ranking	#17 in MvnRepository (See Top Artifacts) #2 in Testing Frameworks & Tools
Used By	20,645 artifacts

Maven

Gradle

SBT

Mill

Ivy

Grape

Leiningen

Buildr

Scope: Test

Format: Kotlin

```
// https://mvnrepository.com/artifact/org.junit.jupiter/junit-jupiter-api
testImplementation("org.junit.jupiter:junit-jupiter-api:6.1.0-M1")
```

Getting Started

Updating Individual Dependencies (contd.)

← → ↺

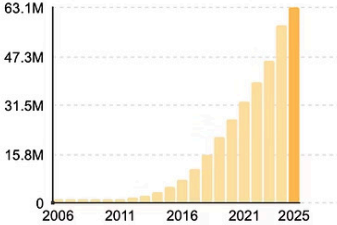
mvnrepository.com/artifact/org.junit.jupiter/junit-jupiter-engine/6.1.0-M1

MVN REPOSITORY

Search for groups, artifacts, categories

Search

Indexed Artifacts (63.1M)



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Logging Bridges

Dependency Injection

XML Processing

Home » org.junit.jupiter » junit-jupiter-engine » 6.1.0-M1



JUnit Jupiter Engine » 6.1.0-M1

Core package for the JUnit Jupiter test engine.

License	EPL 2.0
Categories	Testing Frameworks & Tools
Tags	quality junit testing engine
HomePage	https://junit.org/ Inspect URL
Date	Nov 17, 2025
Files	pom (3 KB) jar (349 KB) View All
Repositories	Central
Ranking	#21 in MvnRepository (See Top Artifacts) #3 in Testing Frameworks & Tools
Used By	19,538 artifacts

Maven

Gradle

SBT

Mill

Ivy

Grape

Leiningen

Buildr

Scope: Test

Format: Kotlin

// https://mvnrepository.com/artifact/org.junit.jupiter/junit-jupiter-engine

testImplementation("org.junit.jupiter:junit-jupiter-engine:6.1.0-M1")

Getting Started

Updating Individual Dependencies (contd.)

← → ↺

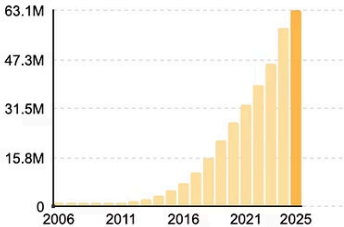
mvnrepository.com/artifact/org.junit.jupiter/junit-jupiter-params/6.1.0-M1

MVN REPOSITORY

Search for groups, artifacts, categories

Search

Indexed Artifacts (63.1M)



Year	Indexed Artifacts (M)
2006	0
2011	0.5
2016	1.5
2021	15.8
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Annotation Libraries

HTTP Clients

Language Runtime

Logging Bridges

Dependency Injection

XML Processing

Home » org.junit.jupiter » junit-jupiter-params » 6.1.0-M1



JUnit Jupiter Params » 6.1.0-M1

JUnit Jupiter extension for running parameterized tests.

License	EPL 2.0
Categories	Testing Frameworks & Tools
Tags	quality junit testing
HomePage	https://junit.org/ Inspect URL
Date	Nov 17, 2025
Files	pom (3 KB) jar (289 KB) View All
Repositories	Central
Ranking	#76 in MvnRepository (See Top Artifacts) #10 in Testing Frameworks & Tools
Used By	8,384 artifacts

Maven

Gradle

SBT

Mill

Ivy

Grape

Leiningen

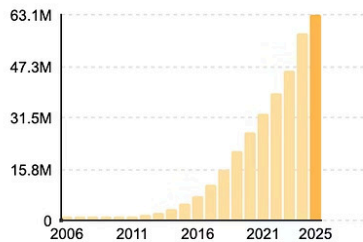
Buildr

Scope: Test

Format: Kotlin

```
// https://mvnrepository.com/artifact/org.junit.jupiter/junit-jupiter-params
testImplementation("org.junit.jupiter:junit-jupiter-params:6.1.0-M1")
```

Indexed Artifacts (63.1M)



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Logging Frameworks

Java Specifications

JSON Libraries

Core Utilities

Mocking

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HTTP Clients

Language Runtime

Logging Bridges

Dependency Injection

XML Processing

Home » org.junit.platform » junit-platform-launcher » 6.1.0-M1



JUnit Platform Launcher » 6.1.0-M1

Public API for configuring and launching test plans.

License	EPL 2.0
Categories	Testing Frameworks & Tools
Tags	quality junit testing platform launcher
HomePage	https://junit.org/ Inspect URL
Date	Nov 17, 2025
Files	pom (3 KB) jar (240 KB) View All
Repositories	Central
Ranking	#163 in MvnRepository (See Top Artifacts) #17 in Testing Frameworks & Tools
Used By	3,619 artifacts

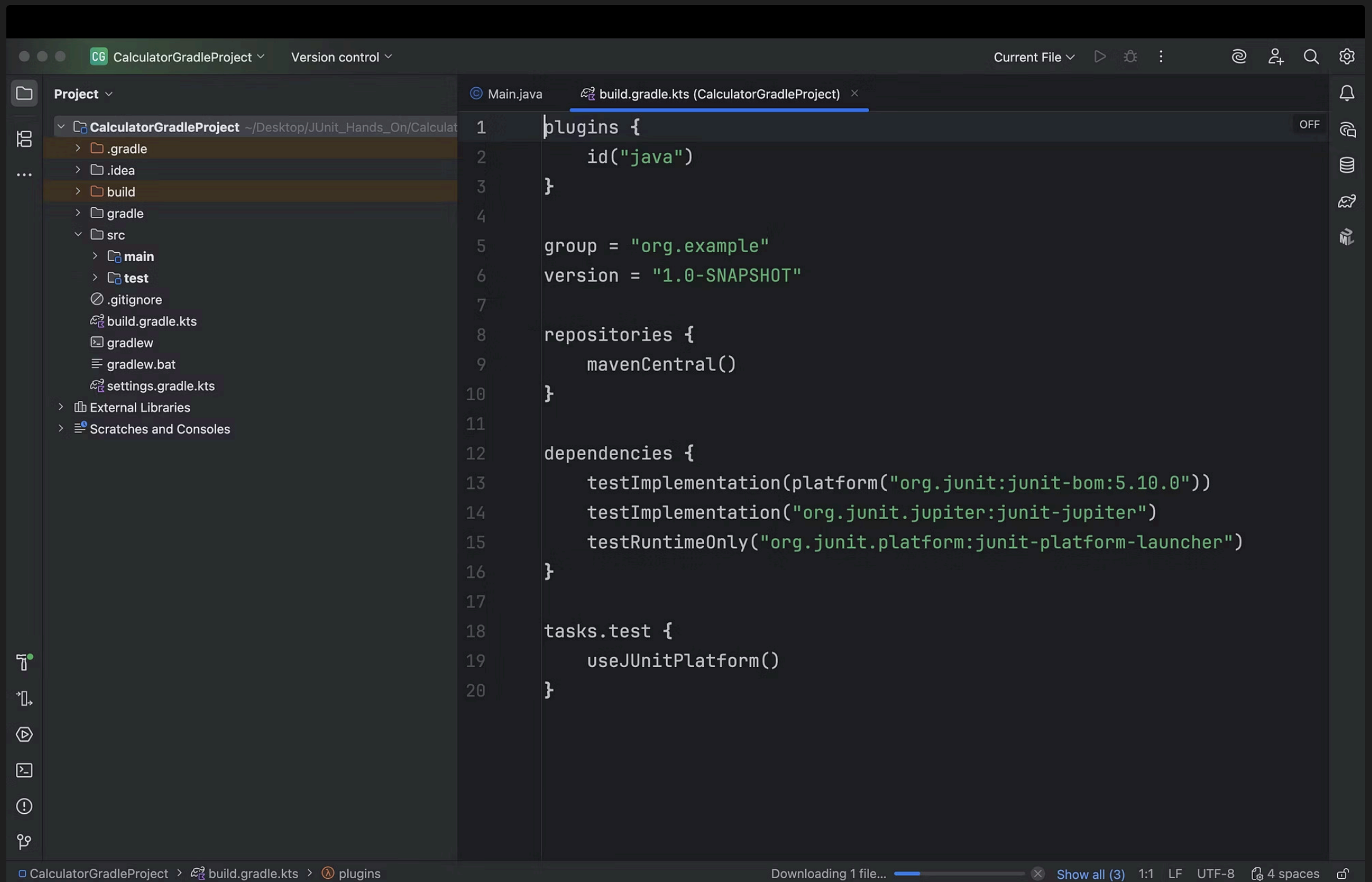
[Maven](#) [Gradle](#) [SBT](#) [Mill](#) [Ivy](#) [Grape](#) [Leiningen](#) [Buildr](#)Scope: [Test](#) Format: [Kotlin](#)

```
// https://mvnrepository.com/artifact/org.junit.platform/junit-platform-launcher
testImplementation("org.junit.platform:junit-platform-launcher:6.1.0-M1")
```

```
// Required with Gradle 9+
testRuntimeOnly("org.junit.platform:junit-platform-launcher")
```

Getting Started

Updating Individual Dependencies (contd.)

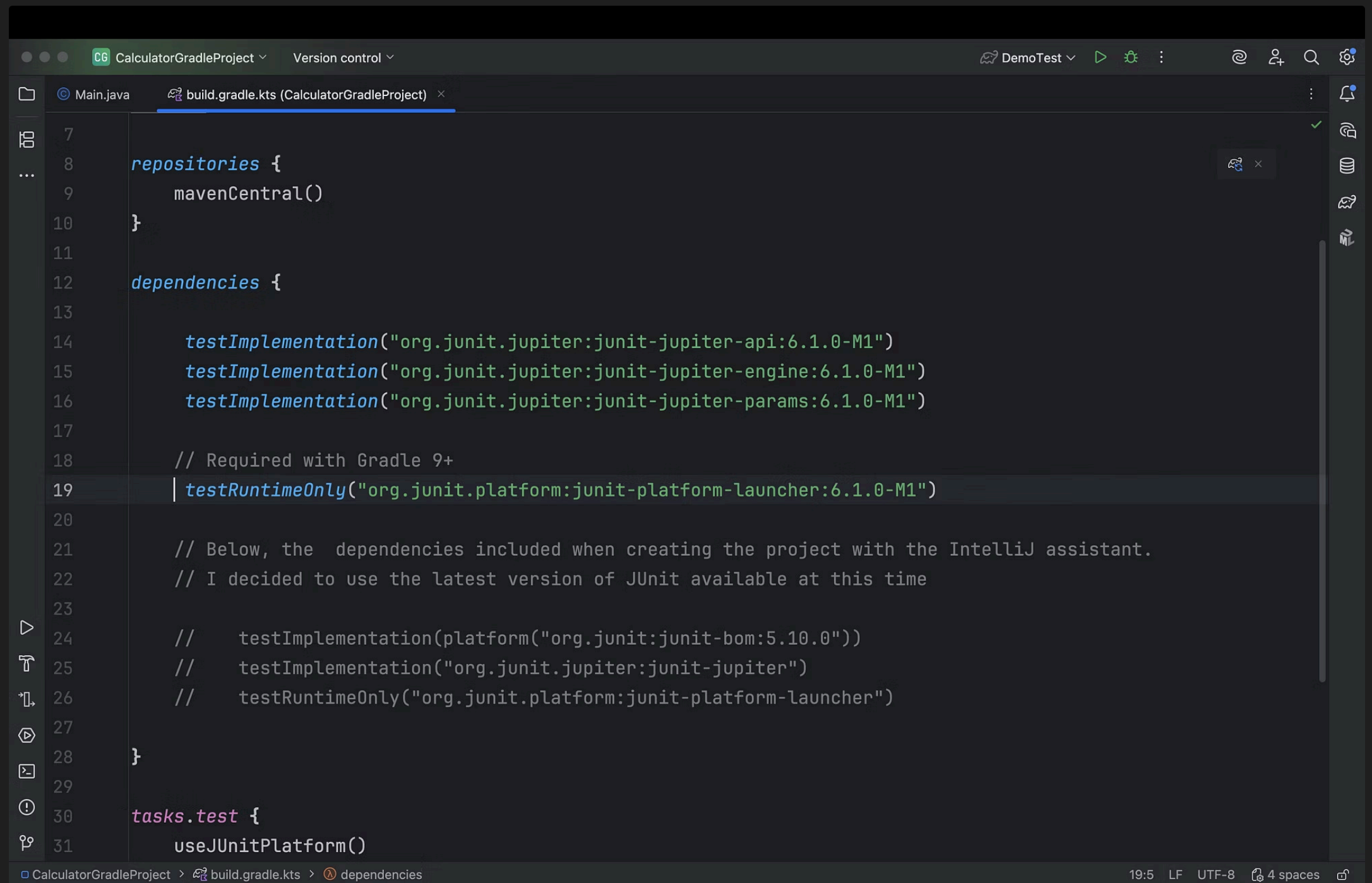


```
1 plugins {
2     id("java")
3 }
4
5 group = "org.example"
6 version = "1.0-SNAPSHOT"
7
8 repositories {
9     mavenCentral()
10 }
11
12 dependencies {
13     testImplementation(platform("org.junit:junit-bom:5.10.0"))
14     testImplementation("org.junit.jupiter:junit-jupiter")
15     testRuntimeOnly("org.junit.platform:junit-platform-launcher")
16 }
17
18 tasks.test {
19     useJUnitPlatform()
20 }
```

Let's **comment out** the **3 dependencies** generated by **default** and **insert** the **newer versions**!

Getting Started

Updating Individual Dependencies (contd.)



```
7
8 repositories {
9     mavenCentral()
10 }
11
12 dependencies {
13
14     testImplementation("org.junit.jupiter:junit-jupiter-api:6.1.0-M1")
15     testImplementation("org.junit.jupiter:junit-jupiter-engine:6.1.0-M1")
16     testImplementation("org.junit.jupiter:junit-jupiter-params:6.1.0-M1")
17
18     // Required with Gradle 9+
19     testRuntimeOnly("org.junit.platform:junit-platform-launcher:6.1.0-M1")
20
21     // Below, the dependencies included when creating the project with the IntelliJ assistant.
22     // I decided to use the latest version of JUnit available at this time
23
24     // testImplementation(platform("org.junit:junit-bom:5.10.0"))
25     // testImplementation("org.junit.jupiter:junit-jupiter")
26     // testRuntimeOnly("org.junit.platform:junit-platform-launcher")
27
28 }
29
30 tasks.test {
31     useJUnitPlatform()
```

```
// Required with Gradle 9+
testRuntimeOnly("org.junit.platform:junit-platform-launcher:6.1.0-M1")
```

Don't forget to **reload** the `build.gradle` file!

Using the JUnit Jupiter Aggregator

Using the JUnit Jupiter Aggregator

- Instead of three separate dependencies, you can add one **aggregator** dependency
- The aggregator includes **API**, **Engine**, and **Params** modules together
- Simplifies configuration and ensures consistent version alignment
- **Optional**: enable test output visibility by adding:
 - `testLogging { showStandardStreams = true }`
- **Reload** the **Gradle** project to apply the new configuration



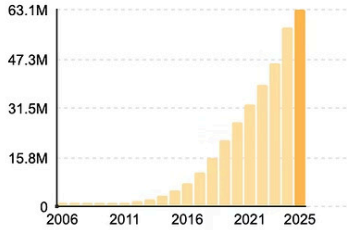
Adding the JUnit 6 Aggregator Dependency to a Gradle Project

mvnrepository.com/artifact/org.junit.jupiter/junit-jupiter/6.1.0-M1

MVN REPOSITORY

Search for groups, artifacts, categories

Indexed Artifacts (63.1M)



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- Mocking
- Web Assets
- Annotation Libraries
- HTTP Clients
- Language Runtime
- Logging Bridges
- Dependency Injection
- XML Processing

Home » org.junit.jupiter » junit-jupiter » 6.1.0-M1

JUnit Jupiter (Aggregator) » 6.1.0-M1

Module "junit-jupiter" of JUnit

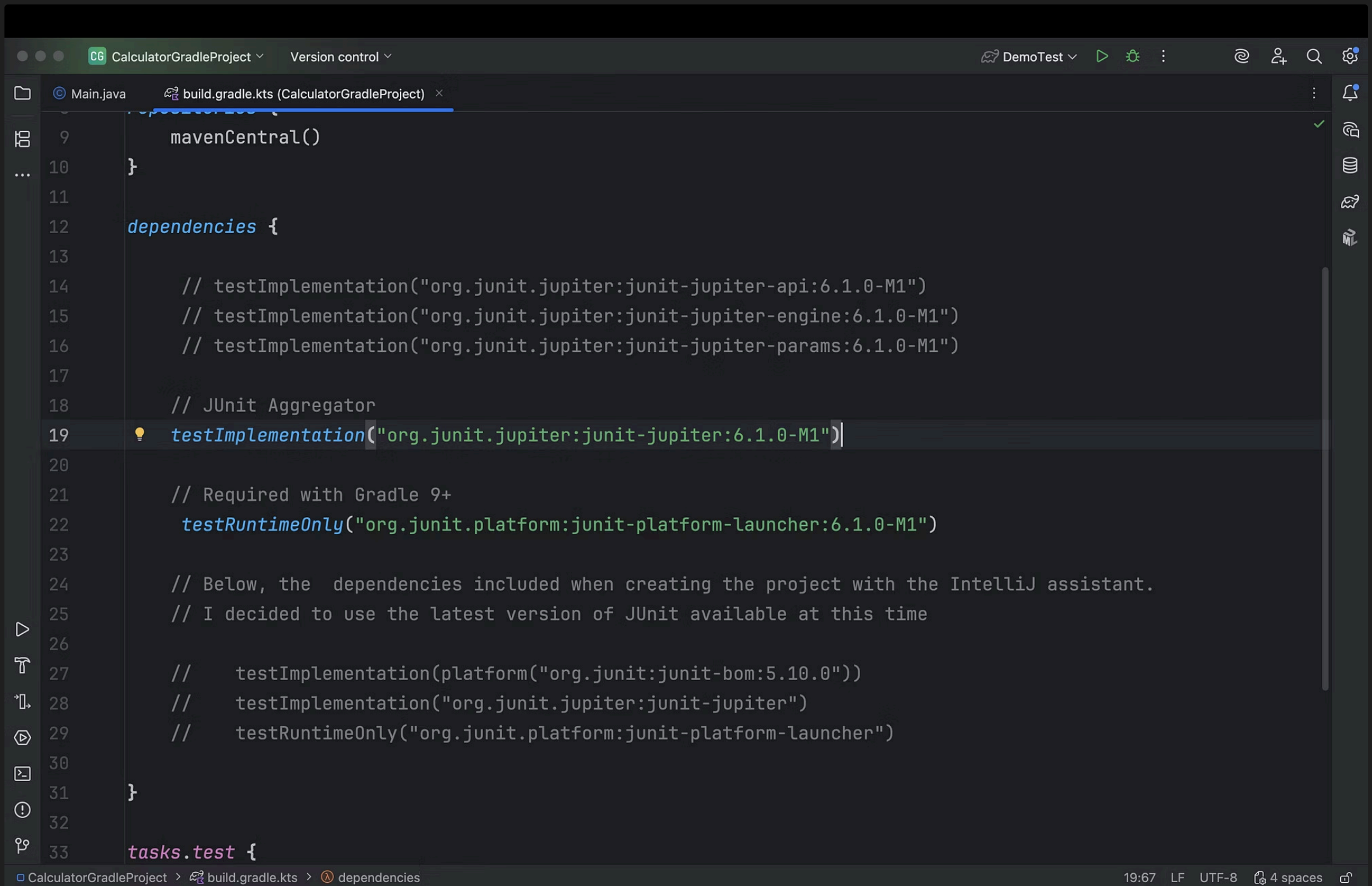
License	EPL 2.0
Categories	Testing Frameworks & Tools
Tags	testing junit quality
HomePage	https://junit.org/ Inspect URL
Date	Nov 18, 2025
Files	pom (3 KB) jar (6 KB) View All
Repositories	Central
Ranking	#53 in MvnRepository (See Top Artifacts) #6 in Testing Frameworks & Tools
Used By	13,190 artifacts

Maven Gradle SBT Mill Ivy Grape Leiningen Buildr

Scope: Test Format: Kotlin

```
// https://mvnrepository.com/artifact/org.junit.jupiter/junit-jupiter
testImplementation("org.junit.jupiter:junit-jupiter:6.1.0-M1")
```

Adding the JUnit 6 Aggregator Dependency to a Gradle Project (contd.)



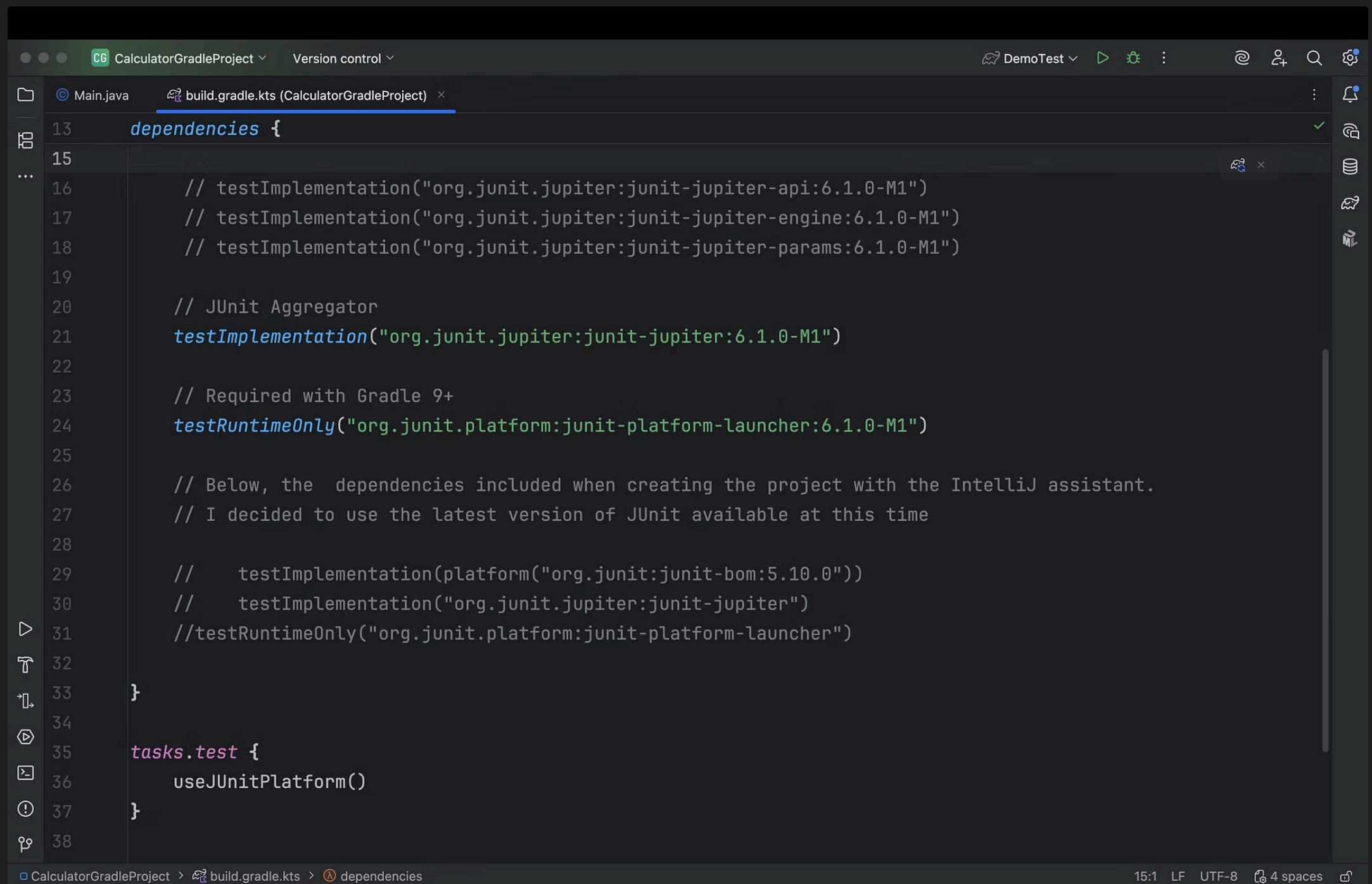
The screenshot shows an IDE window for a project named 'CalculatorGradleProject'. The file 'build.gradle.kts' is open, showing the 'dependencies' block. The code is as follows:

```
9      mavenCentral()
10  }
11
12  dependencies {
13
14      // testImplementation("org.junit.jupiter:junit-jupiter-api:6.1.0-M1")
15      // testImplementation("org.junit.jupiter:junit-jupiter-engine:6.1.0-M1")
16      // testImplementation("org.junit.jupiter:junit-jupiter-params:6.1.0-M1")
17
18      // JUnit Aggregator
19      testImplementation("org.junit.jupiter:junit-jupiter:6.1.0-M1")
20
21      // Required with Gradle 9+
22      testRuntimeOnly("org.junit.platform:junit-platform-launcher:6.1.0-M1")
23
24      // Below, the dependencies included when creating the project with the IntelliJ assistant.
25      // I decided to use the latest version of JUnit available at this time
26
27      // testImplementation(platform("org.junit:junit-bom:5.10.0"))
28      // testImplementation("org.junit.jupiter:junit-jupiter")
29      // testRuntimeOnly("org.junit.platform:junit-platform-launcher")
30
31  }
32
33  tasks.test {
```

The IDE interface includes a top toolbar with icons for running tests, a left sidebar with project navigation, and a bottom status bar showing the current file path: 'CalculatorGradleProject > build.gradle.kts > dependencies'.

Adding the JUnit 6 Aggregator Dependency to a Gradle Project (contd.)

Reload the `build.gradle` file.

A screenshot of an IDE window showing the `build.gradle.kts` file for a project named `CalculatorGradleProject`. The file is open in a tab, and the `dependencies` block is visible. The code includes comments and Kotlin DSL syntax for adding JUnit 6 dependencies. The IDE interface includes a top toolbar with icons for running, testing, and other actions, and a bottom status bar showing the current file path and encoding.

```
13 dependencies {
14
15     // testImplementation("org.junit.jupiter:junit-jupiter-api:6.1.0-M1")
16     // testImplementation("org.junit.jupiter:junit-jupiter-engine:6.1.0-M1")
17     // testImplementation("org.junit.jupiter:junit-jupiter-params:6.1.0-M1")
18
19     // JUnit Aggregator
20     testImplementation("org.junit.jupiter:junit-jupiter:6.1.0-M1")
21
22     // Required with Gradle 9+
23     testRuntimeOnly("org.junit.platform:junit-platform-launcher:6.1.0-M1")
24
25     // Below, the dependencies included when creating the project with the IntelliJ assistant.
26     // I decided to use the latest version of JUnit available at this time
27
28     //     testImplementation(platform("org.junit:junit-bom:5.10.0"))
29     //     testImplementation("org.junit.jupiter:junit-jupiter")
30     //testRuntimeOnly("org.junit.platform:junit-platform-launcher")
31
32 }
33
34 tasks.test {
35     useJUnitPlatform()
36 }
37
38
```

Conclusion

- **Gradle** offers flexible dependency management for **JUnit 6**
- You may configure tests using **individual components** or a **single aggregator** dependency
- **Maven Repository** remains the primary source for updated dependency versions
- Additional **logging** improves debugging during test execution



Open Floor Discussion

Do you prefer granular control or an aggregated approach for test dependencies?

How often do you manually update dependency versions in your Gradle builds?

Have you used parameterized tests with JUnit Jupiter before?